

Design and Evaluation of Lattice Structures for Regenerative Cooling in Additively Manufactured Rocket Nozzles

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Abstract

Regenerative cooling is used in modern rocket nozzles to prevent thermal failure, but current conventional channel-based designs face thermal and structural limitations. High pressure environments result in low-cycle fatigue, wall thinning, and microcrack formation, ultimately reducing nozzle lifespan and reusability. This research explores the use of lattice structures as an alternative to conventional cooling channels to improve heat transfer, and structural durability. Using additive manufacturing, body-centered cubic (BCC) lattice structures will be designed and analyzed. SolidWorks in addition to Finite Element Analysis (FEA) in ANSYS, were used to evaluate stress distribution and thermal performance. Results demonstrated that increase in lattice rod thickness reduced the maximum wall temperature, stress, and strain but led to high pressure drop through the cooling channels. Comparisons with conventional rectangular cooling channels validated the analysis procedure and highlighted the trade-off between cooling performance and flow resistance in lattice designs. Overall, the study confirmed the potential of lattice structures to improve rocket nozzle durability and reusability in reusable launch vehicles, while also identifying design challenges that must be addressed for practical implementation.

Background

Since the Space Race Era, we are finally seeing the advancement of rocket propulsion again, due to interest in the private sector. From a private investor perspective, if we can make rockets reusable, we can make multiplanetary travel monetarily feasible. Therefore, it is important to focus on the reusability and efficiency of rocket nozzles, a key component of rocketry. Specifically, we must improve their thermal efficiency and life span.

Table 1:RocketV1 Prototype

Design Parameters		CEA Analysis	
Thrust	100N	T_c (K)	2010.7318
O/F	1.3	T_t (K)	1753.1925
Pc	2.98 MPa	T_e (K)	1038.8894
Pe	.1014 Mpa		

• NASA CEA Analysis [6] was used to to find the rocket nozzle temperatures in the chamber (T_c), throat (T_t), and exit nozzle (T_e).

• Stainless Steel has a softening temperature of 1050 K, therefore regenerative cooling in rocket nozzles is necessary

ROLE OF LATTICE STRUCTURES IN COOLING CHANNEL DESIGN

- Conventional channel-type regenerative cooling systems suffer due to structural and thermal limitations; The Doghouse Effect
- Lattice structures improve cooling efficiency by enhancing heat transfer and significantly reducing the weight of the nozzle.
- (BCC) lattice structures allow for even load distribution, reducing damage to the inner wall, and in turn increasing the lifespan of the nozzle by 359% [3].

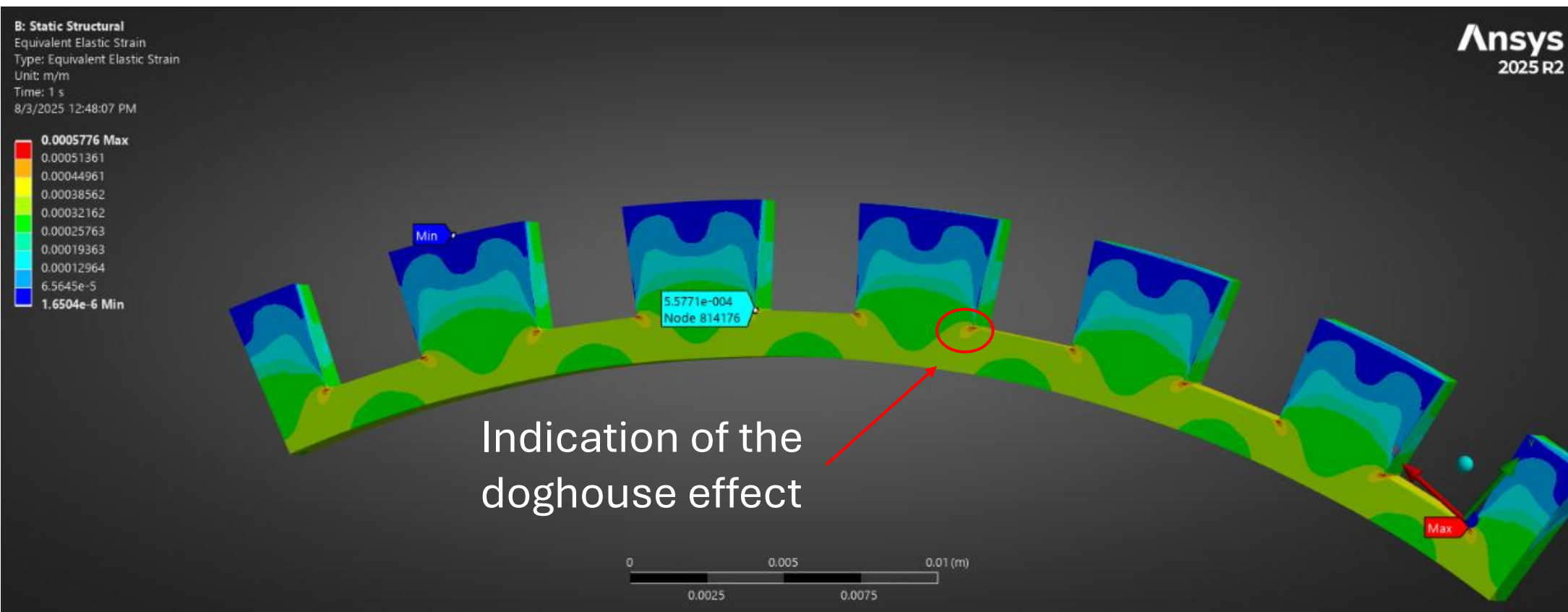


Figure 1:Rectangular Cooling Stress Plot

Methodology

An Unconventional Approach was taken to model the lattice structures using solid works

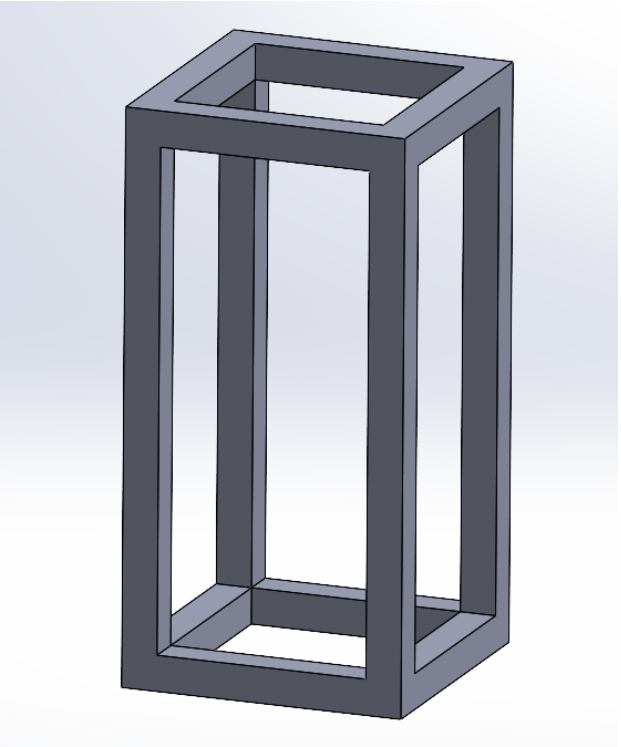


Figure 1: Box created using global variables

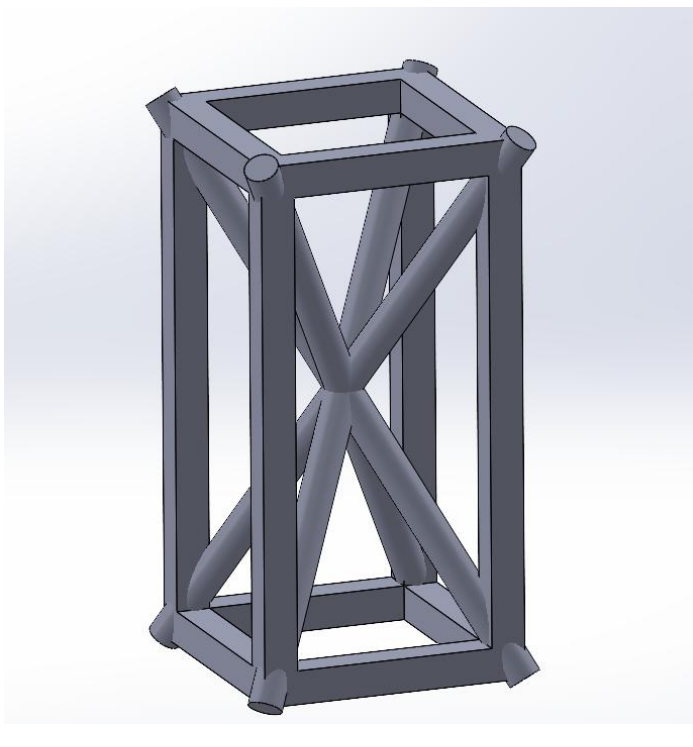


Figure 2: Reference planes used to extrude lattice structure

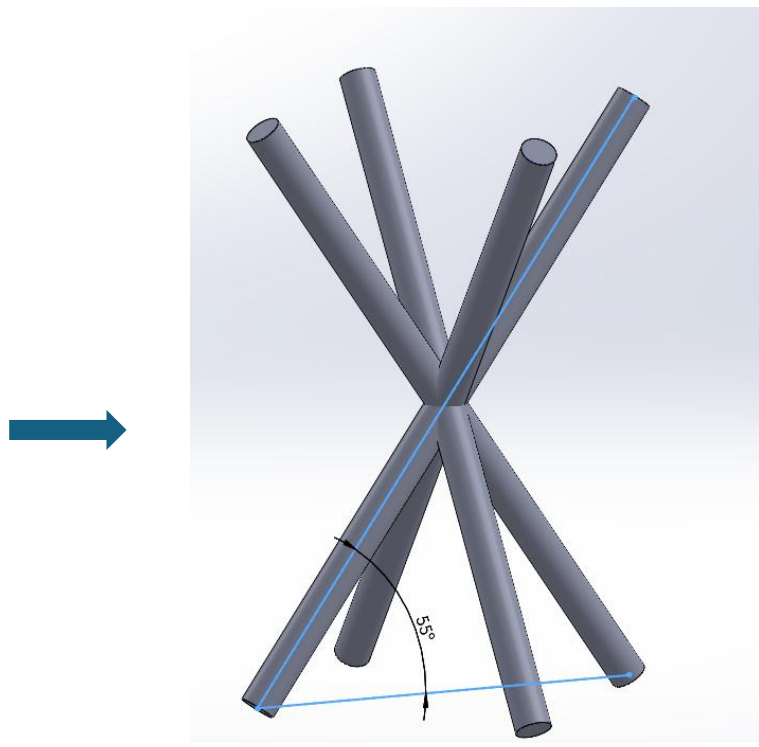


Figure 3: Box removed

To validate analysis procedures and ensure compatibility with published results [7], the material and geometry specifications for the combustion chamber segment was modeled as a cylinder of diameter 77.8 mm and length of 10mm. Stainless-Steel-310-annealed was selected for the nozzle jacket, and Copper-high-conductivity-cast was selected for the combustion side wall and rectangular/lattice cooling channels

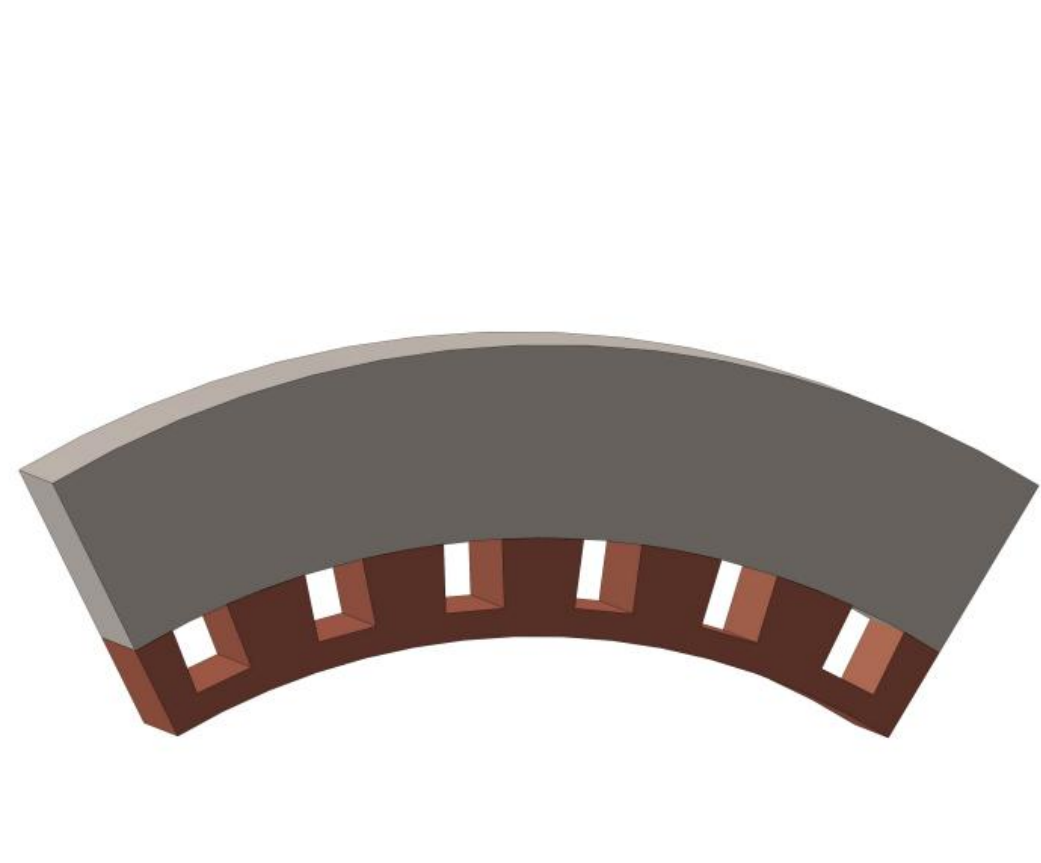


Figure 4: Regular Cooling Channels

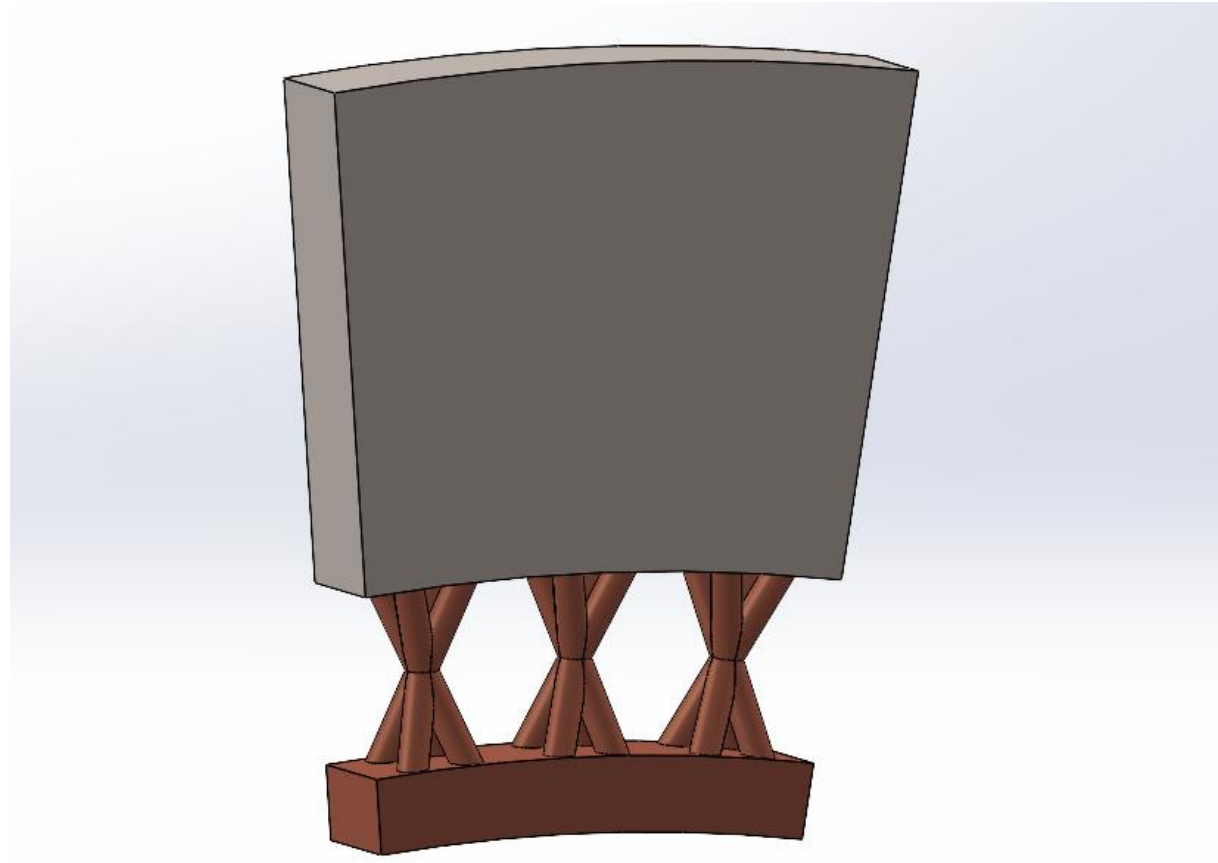


Figure 5: Cooling Channels Lattice 55-.4

In Ansys Fluent, water was chosen to simply the model. Fluent results were exported to ANSYS Static Structural for thermos structural analysis. The following conditions were made:

Table 2: Boundary Conditions

ANSYS FLUENT		ANSYS STATIC STRUCTURAL	
Mass Flow Inlet	.882 $\frac{kg}{s}$ at 290K	Combustion Pressure	2.98 MPa
Pressure Outlet	290K	Channels Pressure	3.15 MPa
Combustion Wall	$q'' = 12 \frac{MW}{m^2}$ t = 1.5mm		

Results

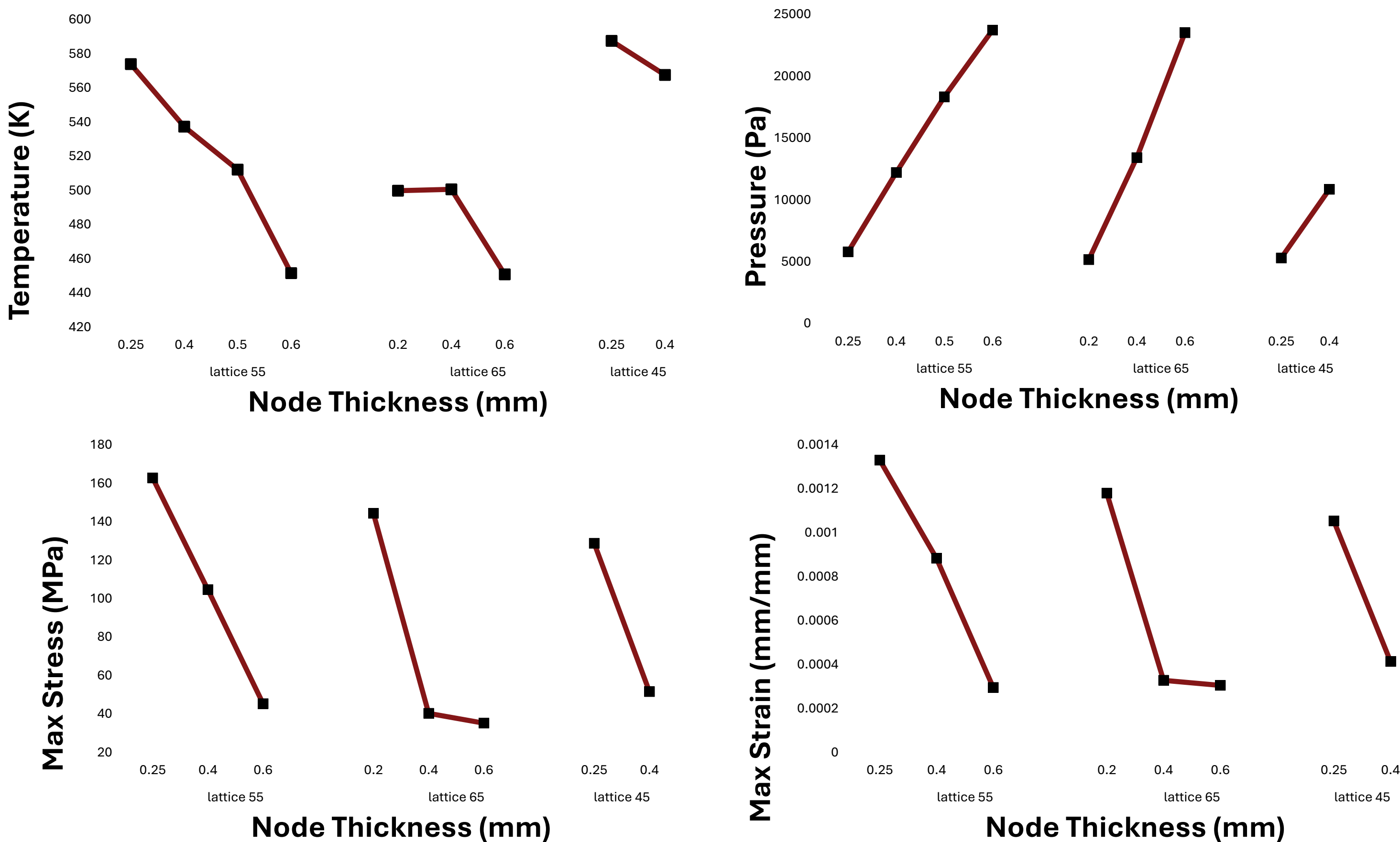


Table 3: Raw Data

	Max Temp (°K)	Max Node Stress (Mpa)	Max Node Strain (m/m)	ΔP (Pa)
rectangular	498.8	67.791	0.0005776	4080.486
lattice 55-.25	574	162.74	0.0013304	5792.1
lattice 55-.4	537.4	104.66	0.00088362	12209.63
lattice 55-.5	512.2	---	---	18326.48
lattice 55-.6	451.5	45.248	0.00029483	23718.84
lattice 55-.8	362.1	---	----	676550.5
lattice 65-.2	499.8	144.36	0.0011804	5165.414
lattice 65-.4	500.6	40.276	0.00032733	13415.27
lattice 65-.6	450.8	35.151	0.00030449	23507.87
lattice45-.25	587.5	128.79	0.0010532	5302.199
lattice45-.4	567.6	51.535	0.00041312	10853.17

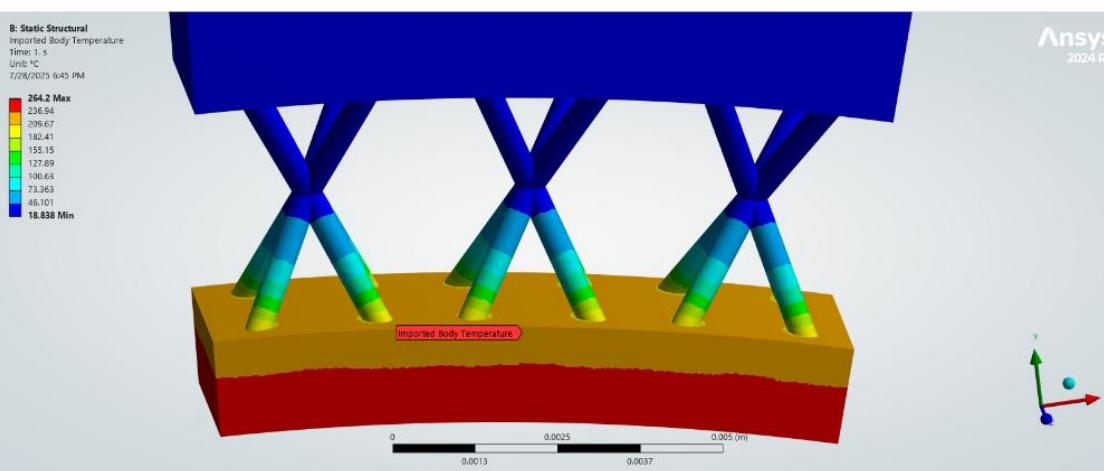


Figure 6: 55° Lattice .4mm thickness Temp Plot

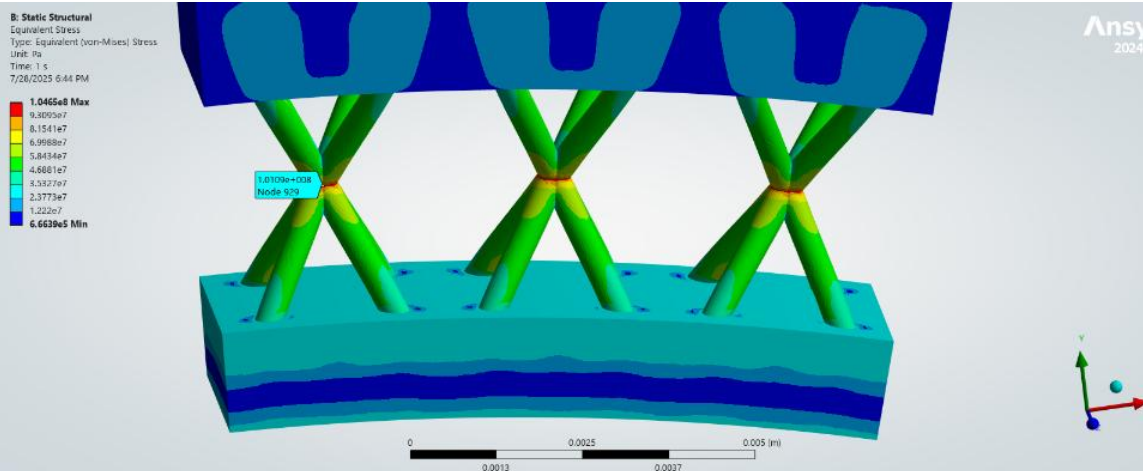


Figure 7: 55° Lattice .4mm thickness stress plot

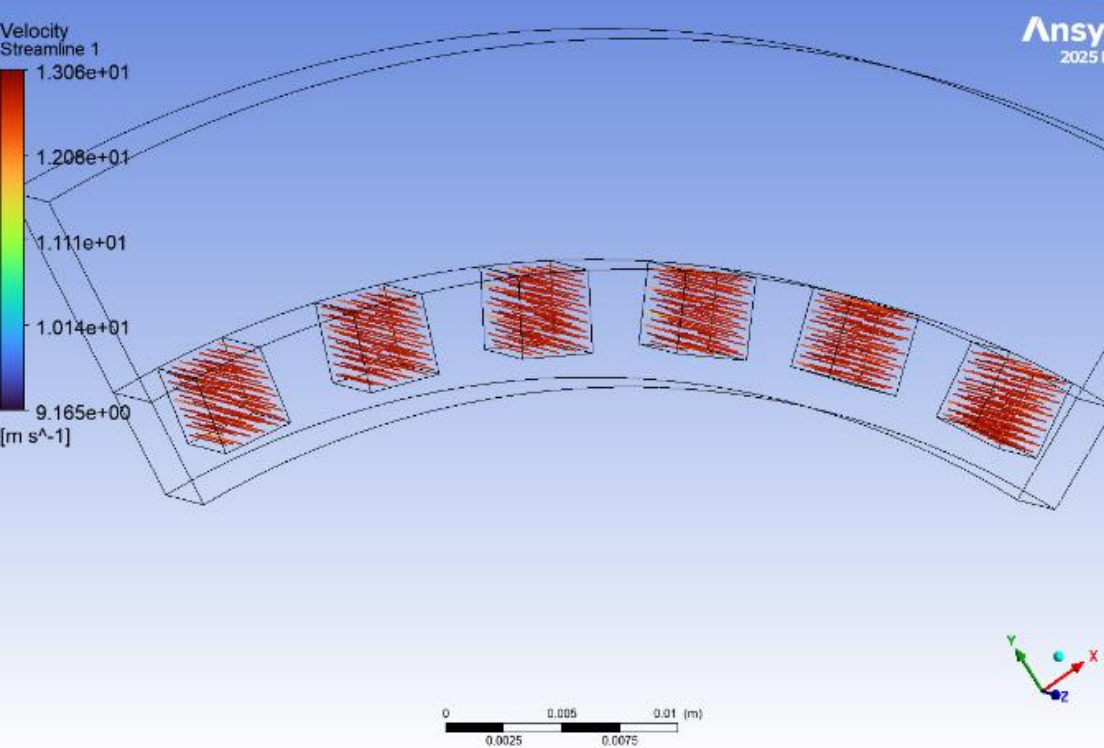


Figure 8: Rectangular Channels Velocity Streamline

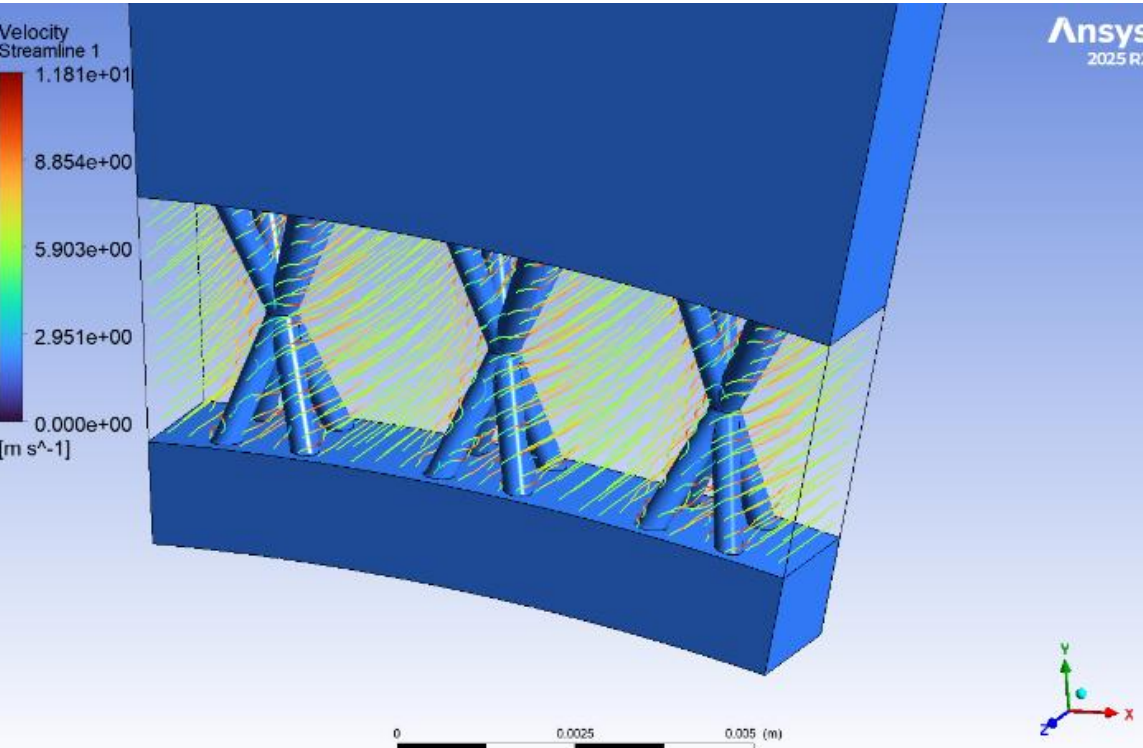


Figure 9: 65° Lattice .4mm Velocity Streamline

Rectangular channels maintain a uniform velocity distribution, with smooth and parallel streamlines throughout the whole channel, indicating minimal velocity disturbance, and efficient fluid flow. In contrast, for the lattice channels, streamlines must bend around the lattice rods, producing regions of disturbed flow and significant changes in velocity. These effects indicate flow resistance and turbulence, which in turn enhance wall-fluid interactions and improve heat transfer. However, this benefit comes at the cost of a substantially higher overall pressure drop across the lattice structure.

Conclusion

In general, the following conclusions can be made:

- As node thickness increases Temperature, Max Stress, Max Strain and weight of the nozzle decreases
- As node thickness increases, Pressure Drop in the Cooling Channel greatly increases
- As the angle increases, weight and maximum temperature decreases
- As the angle increases pressure drop increases

References

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